Hypertensive Emergencies

Hypertensive Emergency (Hypertensive Crisis)

- definition: severe elevation of BP with evidence of end-organ damage (CNS, retinal, CVS, renal, GI)
- etiology
  - essential HTN, emotional exertion, pain, use of sympathomimetic drugs (cocaine, amphetamine, etc.), MAOI use with ingestion of tyramine-containing food (cheese, red wine, etc.), pheochromocytoma, pregnancy
- clinical feature

Table 21. Signs and Symptoms of Hypertensive Emergencies

	CNS	Retinal	Renal	Cardiovascular	Gastrointestinal
Complication	Stroke/TIA, headache, altered mental status, seizures, hemorrhage	Vision change, hemorrhage, exudates, papilledema	Nocturia, elevated Cr, proteinuria, hematuria, oliguria	Ischemia/angina, infarction, dissection (back pain), CHF	N/V, abdominal pain, elevated liver enzymes

HELLP Syndrome (seen only in preeclampsia/eclampsia)
Hemolytic anemia
Elevated **L**iver enzymes
Low **P**latelet count

- investigations
  - blood work: CBC, electrolytes, BUN, Cr
  - urinalysis
  - peripheral blood smear: to detect microangiopathic hemolytic anemia
  - CXR: if SOB or chest pain
  - ECG, troponins, CK: if chest pain
  - CT head: if neurological findings or severe headache
  - toxicology screen if sympathomimetic overdose suspected
- management
  - in general, strategy is to gradually and progressively reduce BP in 24-48 h
  - lower BP by 25% over the initial 60 min by initiating antihypertensive therapy (usually nitroprusside and labetatol)
  - if preeclampsia, immediately consult OB/GYN (see Obstetrics, OB24)
  - establish arterial line; transfer to ICU for further reduction in BP under monitored setting
  - in case of ischemic stroke: do no rapidly reduce BP, maintain BP >150/100 for 5 d
  - in case of aortic dissection: rapid reduction of sBP to 110-120 STAT (do not resuscitate with IV fluids)
  - in case of excessive catecholamines: avoid β-blockers (e.g. labetalol)
  - in case of ACS: address ischemia initially, then BP

Catecholamine-Induced Hypertensive Emergencies
Avoid use of non-selective β-blockers as they inhibit β-mediated vasodilation and leave α-adrenergic vasoconstriction unopposed

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Hypertensive Urgency

- definition: severely elevated BP (usually sBP >180, dBP >110) with no evidence of end-organ damage
- most often caused by non-adherence to prescriptions
- treatment: re-initiate antihypertensive therapy, acute BP reduction not indicated
- goal: differentiate hypertensive emergencies from hypertensive urgencies

With CNS manifestations of severe HTN, it is often difficult to differentiate causal relationships (i.e. HTN could be secondary to a cerebral event with an associated Cushing reflex)

Table 22. Commonly Used Agents for the Treatment of Hypertensive Crisis

Drug	Dosage	Onset of Action	Duration of Action	Adverse Effects*	Special Indications
VASODILATORS					
Sodium Nitroprusside (vascular smooth muscle dilator) 1st line	0.25-10 μg/kg/min	Immediate	3-5 min	N/V, muscle twitching, sweating, cyanide intoxication, coronary steal syndrome	Most hypertensive emergencies (especially CHF, aortic dissection) Use in combination with β-blockers (e.g. esmolol) in aortic dissection Caution with high ICP and azotemia
Nicardipine (CCB)	2 mg IV bolus, then 4 mg/kg/h IV	15-30 min	40 min	Tachycardia, headache, flushing, local phlebitis (e.g. encephalopathy, renal failure, eclampsia, sympathetic crisis)	Most hypertensive emergencies Caution with acute CHF
Fenoldopam Mesylate (dopamine receptor antagonist)	0.05-0.1 μm/kg/min IV	<5 min	8-10 min	Tachycardia, headache, nausea, flushing (e.g. acute RF)	Most hypertensive emergencies Caution with glaucoma
Enalapril (ACEI)	0.625-1.25 mg IV q6h	15-30 min	12-24 h	Theoretical fall in pressure in high renin states not seen in studies	Acute LV failure Avoid in acute MI, pregnancy, acute RF
Nitroglycerin	5-20 μg/min IV	1-2 min	3-5 min	Hypotension, bradycardia, headache, lightheadedness, dizziness	MI/pulmonary edema
Hydralazine	5-10 mg IV/IM q20min (max 20 mg)	5-20 min	2-6 h	Dizziness, drowsiness, headache, tachycardia, Na^+ retention	Eclampsia
ADRENERGIC INHIBITORS					
Labetalol	20 mg IV bolus q10min or 0.5-2 mg/min	5-10 min	3-6 h	Vomiting, scalp tingling, burning in throat, dizziness, nausea, heart block, orthostatic hypotension	Most hypertensive emergencies (especially eclampsia) Avoid in acute CHF, heart block >1st degree
Esmolol	250-500 μg/kg/min 1 min, then 50 μg/kg/min for 4 min; repeat	1-2 min	10-20 min	Hypotension, nausea, bronchospasm	Aortic dissection, acute MI SVT dysrhythmias, perioperative HTN Avoid in acute CHF, heart block >1st degree
Phentolamine	5-15 mg q5-15min	1-2 min	3-10 min	Tachycardia, headache, flushing	Catecholamine excess (e.g. pheochromocytoma)

*Hypotension may occur when using any of these agents

Alcohol Related Emergencies

• see Psychiatry, PS26

Acute Intoxication

- slurred speech, CNS depression, disinhibition, lack of coordination
- nystagmus, diplopia, dysarthria, ataxia, may progress to coma
- hypotension (peripheral vasodilation)
- if obtunded, rule out
  - head trauma/intracranial hemorrhage
  - associated depressants, toxic alcohols
- may also contribute to respiratory/cardiac depression
  - hypoglycemia (screen with bedside glucometer)
  - hepatic encephalopathy: confusion, altered LOC, coma
- precipitating factors: GI bleed, infection, sedation, electrolyte abnormalities, protein meal
  - Wernicke's encephalopathy (ataxia, ophthalmoplegia, delirium)
  - post-ictal state, basilar stroke

Alcohol levels correlate poorly with intoxication

Alcohol intoxication may invalidate informed consent

Withdrawal

- beware of withdrawal signs
- treatment
  - diazepam 10-20 mg IV/PO or lorazepam 2-4 mg IV/PO q1h until calm
- frequency of dosing may have to be increased depending on clinical response
  - may use CIWA protocol and give benzodiazepines as above until CIWA <10
  - thiamine 100 mg IM/IV then 50-100 mg/d
  - magnesium sulfate 4 g IV over 1-2 h (if hypomagnesemic)
  - admit patients with delirium tremens or multiple seizures

CIWA Withdrawal Symptoms
- N/V
- Tremor
- Paroxysmal sweats
- Anxiety
- Agitation
- Visual disturbances
- Tactile disturbances
- Auditory disturbances
- Headache
- Disorientation

10 symptoms each scored out of 7 except orientation, which is scored out of 4

Table 35. Alcohol Withdrawal Signs

Time Since Last Drink	Syndrome	Description
6-8 h	Mild withdrawal	Generalized tremor, anxiety, agitation, but no delirium Autonomic hyperactivity (sinus tachycardia), insomnia, N/V
1-2 d	Alcoholic hallucinations	Visual (most common), auditory, and tactile hallucinations Vitals often normal
8 h-2 d	Withdrawal seizures	Typically brief generalized tonic-clonic seizures May have several within a few hours CT head if focal seizures have occurred
3-5 d	DT	5% of untreated withdrawal patients Severely confused state, fluctuating LOC Agitation, insomnia, hallucinations/delusions, tremor Tachycardia, hyperpyrexia, diaphoresis High mortality rate

Cardiovascular Complications

- HTN
- cardiomyopathy: SOB, edema
- dysrhythmias ("holiday heart")
- AFib (most common), atrial flutter, SVT, VTach (especially Torsades if hypomagnesemic/hypokalemic)

Metabolic Abnormalities

- alcoholic ketoacidosis
  - antigen metabolic acidosis, urine ketones, low glucose, and normal osmolality
  - history of chronic alcohol intake with abrupt decrease/cessation
  - malnourished, abdominal pain with N/V
  - treatment: dextrose, thiamine (100 mg IM/IV prior to dextrose), volume repletion (with NS)
  - generally resolves in 12-24 h
- other alcohols
  - ethylene glycol: CNS, CVS, renal findings
  - methanol
    - early: lethargy, confusion
    - late: headache, visual changes, N/V, abdominal pain, tachypnea
  - both ethylene glycol and methanol produce severe metabolic acidosis with anion gap (as the alcohol is metabolized) and osmolar gap (initially after ingestion but before metabolism)
  - EtOH co-ingestion is protective

  - treatment
    - urgent hemodialysis required
    - fomepizole 15 mg/kg IV bolus (treatment of choice) or EtOH 10% IV bolus and infusion to achieve blood level of 22 mmol/L (EtOH loading may be done PO)
    - consider folic acid for methanol, and pyridoxine and thiamine for ethylene glycol – both help reduce conversion to active metabolites
- other abnormalities associated with alcohol: hypomagnesemia, hypophosphatemia, hypocalcemia, hypoglycemia, hypokalemia

Acute Pelvic Pain

Etiology

- gynecological
  - ovaries: ruptured ovarian cysts (most common cause of pelvic pain), ovarian abscess, ovarian torsion (rare, 50% will have ovarian mass)
  - fallopian tubes: salpingitis, tubal abscess, hydrosalpinx
  - uterus: leiomyomas (uterine fibroids) – especially with torsion of a pedunculated fibroid or in a pregnant patient (degeneration), PID, endometriosis
  - other: ectopic pregnancy (ruptured/expanding/leaking), spontaneous abortion (threatened or incomplete), endometriosis and dysmenorrhea, sexual or physical abuse
- non-gynecological (see causes of lower abdominal pain above)

History and Physical Exam

- pain: OPQRST
- associated symptoms: vaginal bleeding, discharge, dyspareunia, bowel and/or bladder symptoms
- pregnancy and sexual history
- vitals
- gynecological exam: assess for cervical motion tenderness/"chandelier sign" (suggests PID)
- abdominal exam

Investigations

- β-hCG for all women of childbearing age
- CBC and differential, electrolytes, glucose, Cr, BUN, G&S, PTT/INR
- urinalysis to rule out urologic causes
- vaginal and cervical swabs for C&S during physical exam
- pelvic and abdominal U/S: evaluate adnexa, thickness of endometrium, pregnancy, free fluid or masses in the pelvis
- Doppler flow studies for ovarian torsion

Management

- general: analgesia, determine if admission and consults are needed
- specific:
  - ovarian cysts
    - unruptured or ruptured, and hemodynamically stable: analgesia and follow-up
    - ruptured with significant hemoperitoneum: may require surgery
  - ovarian torsion: surgical detorsion or removal of ovary
  - uncomplicated leiomyomas, endometriosis, and secondary dysmenorrhea can usually be treated on an outpatient basis, discharge with gynecology follow-up
  - PID: broad spectrum antibiotics

Disposition

- referral: gynecological or obstetrical causes requiring surgical intervention, requiring admission, or oncological in nature
- admission: patients requiring surgery, IV antibiotics/pain management
- discharge: negative workup and resolving symptoms; give clear instructions for appropriate follow-up

Gynecological Causes of Pelvic Pain

- Ovarian cyst
- Dysmenorrhea
- Mittelschmerz
- Endometriosis
- Ovarian torsion
- Uterine fibroids/neoplasm
- Adnexal neoplasm
- PID + cervicitis

U/S is the preferred imaging modality in the assessment of acute pelvic pain

Vaginal Bleeding

- see Gynecology, GY9 and Obstetrics, OB13

Etiology

- pregnant patient
  - 1st/2nd trimester: ectopic pregnancy, abortion (threatened, incomplete, complete, missed, inevitable, septic), molar pregnancy, implantation bleeding, friable cervix (most common cause)
  - 2nd/3rd trimester: placenta previa, placental abruption, premature rupture of membranes, preterm labour
  - other: trauma, bleeding cervical polyp, passing of mucous plug
- postpartum
  - postpartum hemorrhage, uterine inversion, retained placental tissue, endometritis
- non-pregnant patients
  - structural (PALM- polyps, adenomyosis, leiomyoma, malignancies/hyperplasia)
  - non-structural (COEIN - coagulopathy, ovulatory, endometrial, iatrogenic, NYD)

History

- characterize bleeding (frequency, duration, number of pads/tampons, types of pads used, cyclicity)
- pain, if present (OPQRSTUV)
- menstrual history, sexual history, STI history, syncope/pre-syncope, malignancy history, family history, hematological history, cardiac history, abdo history
- details of pregnancy, including gush of fluid and fetal movement (>20 wk)

Physical Exam

- ABC (especially noting postural BP/HR and mucous membranes)
- abdominal examination (signs of peritoneal pathology, tenderness, distension, mass)
- speculum examination (NOT IF 2nd/3rd trimester bleeding; perform only if placenta previa has been ruled out with U/S)
  - look for active bleeding, trauma/anomaly, and cervical dilatation
- bimanual examination (cervical motion tenderness, size of uterus, cervical length/dilatation)
- sterile gloves and speculum if pregnant
- POCUS: rule in intra-uterine or ectopic pregnancy, check for free fluid in pelvis/RUQ/LUQ, consider assessment of fluid responsiveness (intra-hepatic IVC collapsibility, carotid flow measurement)

Investigations

- β-hCG test for all patients with childbearing potential
- CBC, blood and Rh type, quantitative β-hCG, PTT, INR
- type & cross if significant blood loss
- transvaginal U/S (rule out ectopic pregnancy and spontaneous abortion)
- abdominal U/S (rule out placenta previa, fetal demise, or retained products post-partum)

Management

- ABCs
- pulse oximeter and cardiac monitors if unstable
- Rh immune globulin (Rhogam®) for vaginal bleeding in pregnancy and Rh-negative mother
- 1st/2nd trimester pregnancy
  - ectopic pregnancy: definitive treatment with surgery or methotrexate
  - intrauterine pregnancy, no concerns of coexistent ectopic: discharge patient with obstetrics follow-up
  - U/S indeterminate or β-hCG >1000-2000 IU: further workup and/or gynecology consult
  - abortions: if complete, discharge if stable; for all others, consult gynecology
- 2nd/3rd trimester pregnancy
  - placenta previa or placental abruption: obstetrics consult for possible admission
- postpartum
  - manage ABCs: start 2 large bore IV rapid infusion, type & cross 4 units of blood, consult OB/GYN immediately
- non-pregnant
  - if unstable admit to gynecology for IV hormonal therapy, possible D&C
  - non-structural abnormalities
    - tranexamic acid to stabilize clots
    - medroxyprogesterone acetate 10 mg PO OD x 10d, warn patient of a withdrawal bleed
  - stable structural abnormalities (fibroids, polyps, endometrial thickening, adenomyosis), outpatient gynecology referral once stable

Disposition

- decision to admit or discharge should be based on the stability of the patient, as well as the nature of the underlying cause; consult OB/GYN for patients requiring admission
- if patient can be safely discharged, ensure follow-up with family physician or gynecologist
- instruct patient to return to ED for increased bleeding, presyncope

Vaginal bleeding can be life-threatening Always start with ABCs and ensure your patient is stable

Need β-hCG ≥1200 to see intrauterine changes on transvaginal U/S

An ectopic pregnancy can be ruled out by confirming an intrauterine pregnancy by bedside U/S unless the patient is using IVF due to the high risk of heterotopic pregnancy

Vaginal bleeding (and its underlying causes) can be a very distressing event for patients; ensure appropriate support is provided

Seizures

• see Neurology, N18

Definition

- paroxysmal alteration of behaviour and/or EEG changes resulting from abnormal, excessive activity of neurons
- status epilepticus: continuous or intermittent seizure activity for greater than 5 min without regaining consciousness (life threatening)

Categories

- generalized seizure (consciousness always lost): tonic/clonic, absence, myoclonic, atonic
- partial seizure (focal): simple partial, complex partial
- causes: primary seizure disorder, structural (trauma, intracranial hemorrhage, infection, increased ICP), metabolic disturbance (hypo-/hyperglycemia, hypo-/hypernatremia, hypocalcemia, hypomagnesemia, toxins/drugs)
- differential diagnosis: syncope, stroke/TIA, pseudoseizures, migraines, movement disorders, narcolepsy/cataplexy

History and Physical Exam

- history of seizures, identify potential precipitants (illness, alcohol use, sleep deprivation)
- preceding aura, rapid onset, brief duration, alterations in consciousness, tonic-clonic movements, and post-ictal symptoms would suggest a seizure
- common signs include loss of bladder/bowel control, tongue biting, emesis and aspiration
- perform vitals, complete neurologic examination and look for injuries to head, spine, and shoulder (dislocations)

Table 17. Concurrent Investigation and Management of Status Epilepticus

Timing	Steps
Immediate	Protect airway with positioning; intubate if airway compromised or elevated ICP Monitor: vital signs, ECG, oximetry; bedside blood glucose Establish IV access Benzodiazepine - IV lorazepam 2 mg at 2mg/min up to 10 mg or IM midazolam 5mg up to 10mg; repeat at 10 min if ineffective Fluid resuscitation IV dextrose if glucose <60 mg/dL Give 50 mL 50% glucose (preceded by thiamine 100 mg IM in adults) Obtain blood samples for glucose, CBC, electrolytes, Ca^{2+}, Mg^{2+}, toxins, and antiepileptic drug levels; consider prolactin, β-hCG Vasopressor support if sBP <90 or MAP <70 mmHg
Urgent	Establish second IV line, urinary catheter If status persists, phenytoin 20 mg/kg IV at 25-50 mg/min in adults; may give additional 10 mg/kg IV 10 min after loading infusion If seizure resolves, antiepileptic drug still required to prevent recurrence EEG monitoring to evaluate for non-convulsive status epilepticus
Refractory	If status persists after maximum doses above, consult ICU and start one or more of: Phenobarbital 20 mg/kg IV at 50 mg/min Midazolam 0.2 mg/kg IV loading dose and 0.1-0.4 mg/kg/h Propofol 2 mg/kg IV at 2-5 mg/kg/h then loading dose then 2-10 mg/kg/h
Post-Seizure	Investigate underlying cause: consider CT, LP, MRI, intracranial pressure monitoring

Note: All interventions should be done as soon as possible
Adapted from Brophy et al. Guidelines for the Evaluation and Management of Status Epilepticus. *Neurocrit Care* 2012;17:3-23

Minimum Workup in an Adult with 1st Time Seizure
CBC and differential
Electrolytes including Ca^{2+}, Mg^{2+}, PO_4^{3-}
Head CT

If administering phenytoin, patient must be on a cardiac monitor as dysrhythmias and/or hypotension may occur

If IV access is not feasible, midazolam 0.2 mg/kg IM up to 10 mg can be used for initial control of seizure in adults

Disposition

- decision to admit or discharge should be based on the underlying disease process identified
  - if a patient has returned to baseline function and is neurologically intact, then consider discharge with outpatient follow-up
- first-time seizure patients being discharged should be referred to a neurologist for follow-up
- admitted patients should generally have a neurology consult
- patient should not drive until medically cleared (local regulations vary)
  - complete notification form to appropriate authority regarding ability to drive
- warn regarding other safety concerns (e.g. no swimming, bathing children alone, etc.)

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"ABCD$_3$EFG" of Toxicology

- basic axiom of care is symptomatic and supportive treatment
- address underlying problem only once patient is stable

A Airway (consider stabilizing C-spine)
B Breathing
C Circulation
D1 Drugs
  - ACLS as necessary to resuscitate the patient
  - universal antidotes

D2 Draw bloods
D3 Decontamination (decrease absorption)
E Expose (look for specific toxidromes)/Examine the patient
F Full vitals, ECG monitor, Foley, X-rays
G Give specific antidotes and treatments

- reassess
- call Poison Information Centre
- obtain corroborative history from family, bystanders

Principles of Toxicology
4 principles to consider with all ingestions:
- Resuscitation (ABCD$_3$EFG)
- Screening (toxidrome? clinical clues?)
- Decrease absorption of drug
- Increase elimination of drug

D1 – Universal Antidotes

- treatments that will not harm patients and may be essential

Dextrose (glucose)
- give to any patient presenting with altered LOC
- measure blood glucose prior to glucose administration if possible
- adults: 0.5-1.0 g/kg (1-2 mL/kg) IV of D50W
- children: 0.25 g/kg (2-4 mL/kg) IV of D25W

Oxygen
- do not deprive a hypoxic patient of oxygen no matter what the antecedent medical history (i.e. even COPD with CO_2 retention)
- if depression of hypoxic drive, intubate and ventilate
- exception: paraquat or diquat (herbicides) inhalation or ingestion (oxygen radicals increase morbidity)

Naloxone (central μ-receptor competitive antagonist, shorter t1/2 than naltrexone)
- antidote for opioids: administration is both diagnostic and therapeutic (1 min onset of action)
- used for the undifferentiated comatose patient
- loading dose
  - adults
    - response to naloxone can be drastic, so stepwise delivery of initial 2 mg bolus is recommended
    - draw up 2 mg to deliver IV/IM/SL/SC or via ETT (ETT dose = 2-2.5x IV dose)
      - 1st dose 0.4 mg (for a chronic opioid user, the initial dose may be much smaller)
      - if no response, deliver second dose 0.6 mg
      - if still no response, deliver remaining 1 mg
  - child
    - 0.01 mg/kg initial bolus IV/IO/ETT (max 2mg per dose)
    - children over 20 kg can receive naloxone 2 mg IV
  - maintenance dose
    - may be required because half-life of naloxone (30-80 min) is much shorter than many opioids
    - hourly infusion rate at 2/3 of initial dose that allowed patient to be roused

Universal Antidotes

DONT
Dextrose
Oxygen
Naloxone
Thiamine (must give BEFORE dextrose)

Administration of naloxone can cause opioid withdrawal in chronic users:
Minor withdrawal may present as lacrimation, rhinorrhea, diaphoresis, yawning, piloerection, HTN, and tachycardia
Severe withdrawal may present as hot and cold flashes, arthralgias, myalgias, N/V, and abdominal cramps

Thiamine (Vitamin B1)
- 100 mg IV/IM with IV/PO glucose to all patients
- given to prevent/treat Wernicke's encephalopathy
- a necessary cofactor for glucose metabolism (may worsen Wernicke's encephalopathy if glucose given before thiamine), but do not delay glucose if thiamine unavailable
- must assume all undifferentiated comatose patients are at risk

Thiamine is deficient in the malnourished. Consider in patients with alcohol use disorder, anorexia, or malnutrition states